Explainable ML for Road Accident Severity & Black Spots

Predict severity. Locate black spots. Explain every decision.

UK STATS19 · 2.05 M accidents · 2005-2017

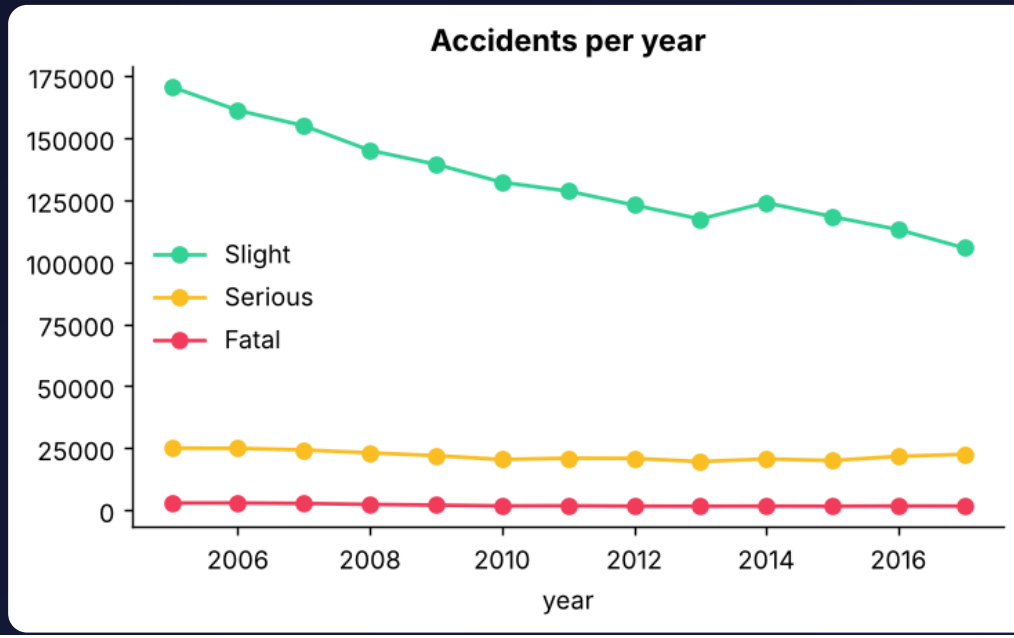
AUCML (TE7483) · AIML · Symbiosis Institute of Technology, Pune

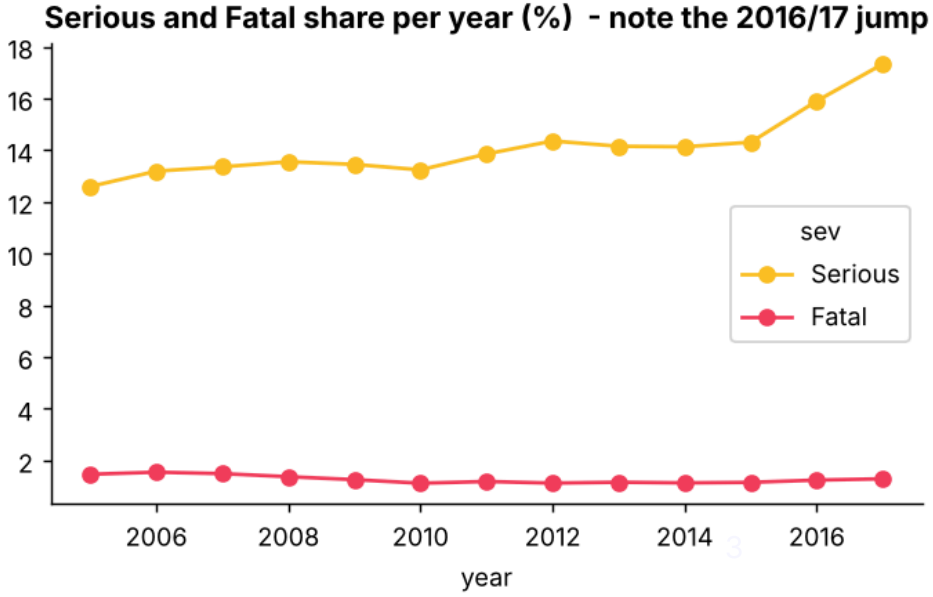
Problem and motivation

- Injury accidents: 85% Slight, 14% Serious, 1.3% Fatal
- Authorities need **where** to act and **why**, with a limited budget
- Gap: accuracy-only reporting, random splits, no link between prediction, hotspots and explanation
- RQ1 drivers of severity · RQ2 black-spot drivers · RQ3 value of leak-free spatial history

Data and leakage control

- 2,046,815 clean rows (0.02% dropped, every step logged)
- Casualty count and police attendance **excluded**: consequences of severity
- Features: scene, time, location and





Pipeline

Clean → Engineer → Leak-free history → **Time split** (train 2006-14 · val 2015 · test 2016-17) → Pipeline (impute + one-hot) → Imbalance strategy → Model → Calibrate + decision scales (on 2015) → Test once → SHAP / LIME / black spots → Web app

Everything fitted on training years only; resampling never touches validation or test.

Models and imbalance study

- Majority, Logistic Regression, Decision Tree, Random Forest, LightGBM, XGBoost
- None · class weights · under-sampling · SMOTE · **tuned decision scales**
- Optuna tuning on validation macro-F1; isotonic calibration; 5-fold CV + paired t-tests
- **Finding:** after scale tuning all strategies land within ~0.01 macro-F1, so the cheap one wins

Results on unseen 2016-17

0.412
macro-F1
(baseline 0.301)

37%
KSI recall

0.698
ROC-AUC

0.025
ECE
(raw 0.137)

Accuracy 72.3% (tuned) · 82.1% (argmax) vs 82.1% majority. Random split: 84.6% vs 84.6%. Binary KSI AUC 0.658.

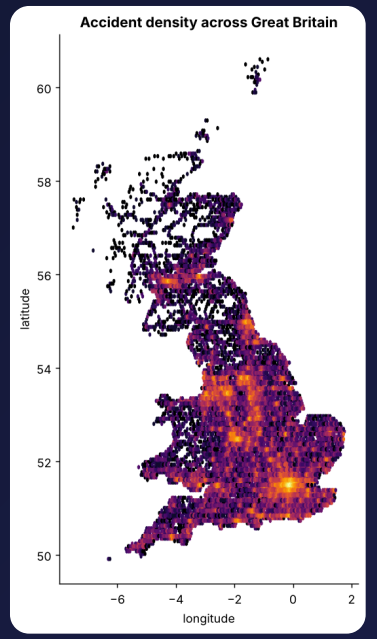
Accuracy above 90% would require leaked features; we report honest, imbalance-aware metrics.

Ablations and honesty

- Removing spatial history, time features or coordinates each lowers macro-F1 and AUC
- Scene-only features cap performance: vehicle and casualty tables would be the next gain
- Distribution shift in 2016-17 (Serious+Fatal 15.5% → 18.6%) handled by calibration on 2015
- Leakage columns shown in the ablation to quantify what they would add

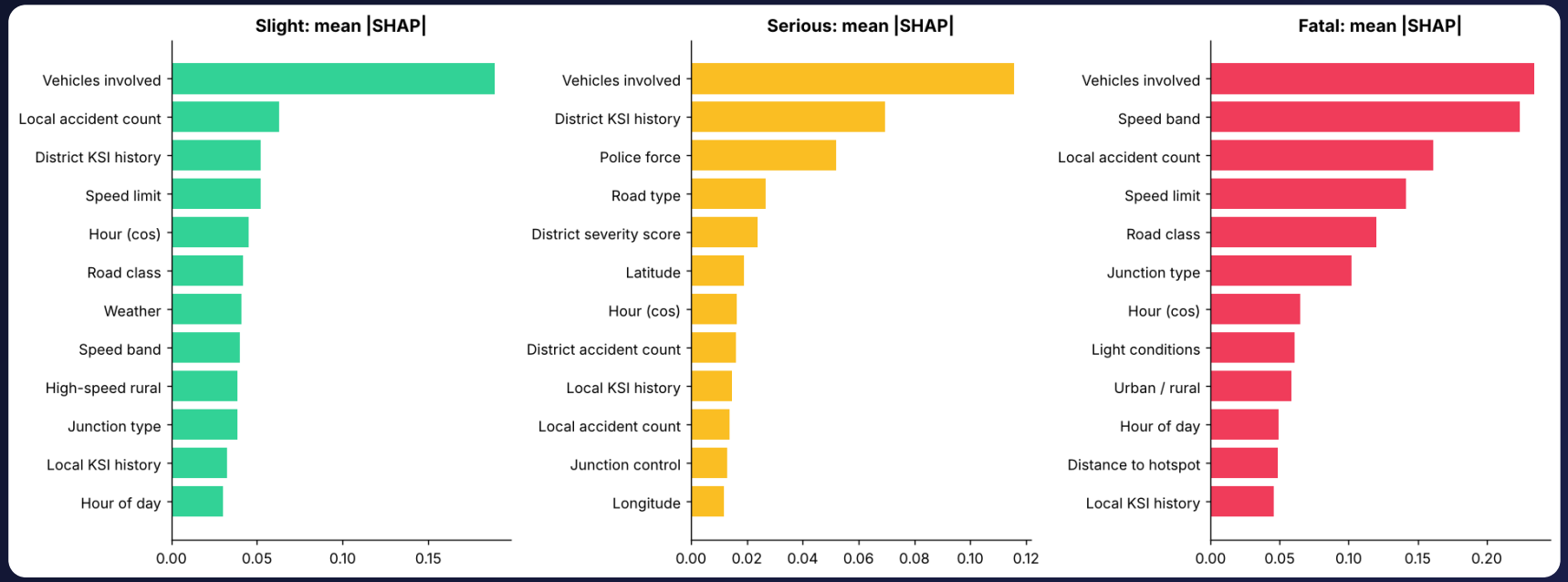
Black-spot detection

- 150 m severity-weighted (EPDO 10/5/1) peak detection; DBSCAN/HDBSCAN rejected as primary (chaining)
- 6,000 sites from 2005-14; **95%** of top-100 persist in 2015-17
- Control sites persist 85%, so we also report rank correlation; sensitivity over radius and weights



Explainability

- SHAP: global, per-class, dependence, local waterfalls, inside vs outside black spots
- LIME compared with SHAP, with seed-stability measured
- Speed limit, local history



Application, limits and future work

- Static React app on GitHub Pages: map predictor, in-browser XGBoost + calibration + exact TreeSHAP, black-spot map, model lab (parity-tested vs Python)
- Limits: UK-only, no traffic exposure, injury-only, scene-level features
- Future: join vehicle/casualty tables, Indian MoRTH data, exposure-normalised risk, spatio-temporal forecasting

Thank you · Questions?